

Finding Marker Genes

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Introduction

Marker genes are genes differentially expressed in one cluster relative to others; they drive cell-type annotation, functional interpretation, and the connection between unsupervised clustering and known biology. Seurat's **FindMarkers** and **FindAllMarkers** implement the standard differential-expression tests (Wilcoxon rank-sum, MAST hurdle, DESeq2) on log-normalised or SCT-transformed data, per cluster, against either all other cells or a specified comparison cluster.

Prerequisites

A working understanding of clustered scRNA-seq output, log-fold-change interpretation, and the difference between cluster-vs-all and pairwise cluster comparisons.

Theory

For each cluster, the marker test compares cells in that cluster to cells in all other clusters (or a specified subset) gene-by-gene. The output for each gene includes:

- **avg_log2FC**: average log-fold change between the target cluster and the reference.
- **pct.1**: fraction of cells in the target cluster expressing the gene.
- **pct.2**: fraction of cells in the reference expressing the gene.
- **p_val_adj**: Benjamini-Hochberg adjusted p -value.

Wilcoxon rank-sum is the default test in Seurat (fast, robust); MAST handles zero-inflation more rigorously via a hurdle model; DESeq2 applies the negative-binomial framework (slow but principled).

Conventional marker criteria: $\log_2 \text{FC} > 0.5$, adjusted $p < 0.05$, and **pct.1** > 0.25 (the gene is expressed in at least a quarter of the cluster's cells).

Assumptions

Cells within a cluster are reasonably homogeneous; the reference set is a valid comparison; multiple testing is corrected across genes within each comparison (but not across the joint cluster set, which would be over-conservative).

R Implementation

```
library(Seurat)

set.seed(2026)
counts <- matrix(rpois(300 * 200, lambda = 2), 200, 300)
rownames(counts) <- paste0("g", 1:200)
colnames(counts) <- paste0("c", 1:300)
# Inject a marker pattern in cells 1-50
counts[1:10, 1:50] <- counts[1:10, 1:50] + 10

so <- CreateSeuratObject(counts)
```

```

so <- NormalizeData(so, verbose = FALSE)
so <- FindVariableFeatures(so, nfeatures = 100, verbose = FALSE)
so <- ScaleData(so, verbose = FALSE)
so <- RunPCA(so, npcs = 10, verbose = FALSE)
so <- FindNeighbors(so, dims = 1:10, verbose = FALSE)
so <- FindClusters(so, resolution = 0.5, verbose = FALSE)

markers_all <- FindAllMarkers(so, only.pos = TRUE, min.pct = 0.25,
                             logfc.threshold = 0.25, verbose = FALSE)
head(markers_all[, c("cluster", "gene", "avg_log2FC", "p_val_adj", "pct.1")])

```

Output & Results

`FindAllMarkers()` returns a data frame with one row per significant marker per cluster, including cluster identity, gene name, log-fold change, adjusted p -value, and percentage of cells expressing in target vs reference. Filtering to `only.pos = TRUE` retains only up-regulated markers, which are typically the most interpretable for cell-type identification.

Interpretation

A reporting sentence: “FindAllMarkers (Wilcoxon test, `min.pct = 0.25`, `logfc.threshold = 0.25`, `only.pos = TRUE`) identified 1{,}240 unique markers across 12 clusters; cluster 2’s top markers (CD8A, GZMB, PRF1) identify cytotoxic T cells, and cluster 5’s markers (MS4A1, CD79A) identify B cells. Annotation followed manual review against the Human Cell Atlas reference.” Always state the test, thresholds, and annotation source.

Practical Tips

- Use `only.pos = TRUE` to retain up-regulated markers; down-regulated markers are less interpretable in scRNA-seq because most genes are dropouts in any given cluster.
- Set `min.pct = 0.25` to filter out genes with very limited expression in the target cluster; this is a quality gate that prevents spurious markers from low-detection genes.
- For publication-quality marker calls, use MAST (`test.use = "MAST"`); it handles zero-inflation more rigorously than Wilcoxon and gives more conservative p -values.
- Always annotate clusters using markers *after* clustering — never adjust clusters to fit pre-conceived markers; the latter is circular reasoning.
- Do not perform DE testing between clusters produced by the same clustering on the same data; this is a form of double-dipping that inflates significance. Use post-hoc validation or independent test data instead.
- For automated cell-type annotation, pair markers with reference-based tools (`SingleR`, `scType`, `Azimuth`); they reduce manual annotation effort substantially.

R Packages Used

`Seurat::FindMarkers()` and `FindAllMarkers()` for the canonical workflow; `MAST` for hurdle-based DE testing; `DESeq2` for negative-binomial DE on pseudobulk; `SingleR` and `Azimuth` for reference-based automated annotation; `presto` for fast Wilcoxon implementations on large datasets.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.

- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with Seurat